

CM0246 Discrete Structures

Functions

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Preliminaries

Textbook

Kenneth H. Rosen ([1988] 2004). Matemática Discreta y sus Aplicaciones.

Convention

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

Preliminaries

Remark

Recall that if an element of a set is listed more than once it doesn't matter.

Example

$$\{1, 3, 3, 3, 5, 5, 5, 5\} = \{1, 3, 5\}.$$

Preliminaries

Notation

$$\mathbb{N} = \{0, 1, 2, \dots\}$$

(natural numbers,
non-negative integers)

$$\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$$

(integers)

$$\mathbb{Z}^+ = \{1, 2, 3, \dots\}$$

(positive integers)

$$\mathbb{Q} = \{p/q \mid p, q \in \mathbb{Z} \text{ and } q \neq 0\}$$

(rational numbers)

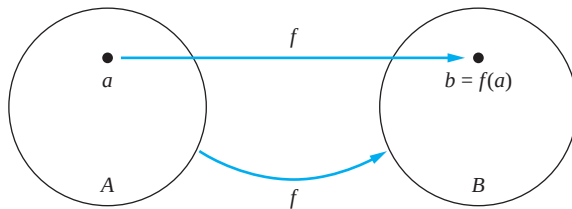
$$\mathbb{R} = (-\infty, \infty)$$

(real numbers)

Functions

Definition

Let A and B be sets. A **function** (**map**, **mapping** or **transformation**) f from A to B , denoted $f : A \rightarrow B$, is an assignment of **exactly** one element of B to **each** element of A .[†]



[†]Figure source: (Rosen [1988] 2012, § 2.3, Fig. 2).

Functions

Specification of functions

- Explicitly

Functions

Specification of functions

- Explicitly
- Formula

Functions

Specification of functions

- Explicitly
- Formula
- Programming languages

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Specification of functions

- Explicitly
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(Advance) question

Are all the functions define using a programming language really functions?

Functions

Definitions

Let f be a function from A to B :

Domain of f : A

Codomain of f : B

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If $f(a) = b$:

b is the **image** of a

a is a **preimage** of b

The **range** or **image** of f : Set of all images of elements of A

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Definitions

Let f be a function from A to B :

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Codomain of f : B

If $f(a) = b$:

b is the **image** of a

a is a **preimage** of b

The **range** or **image** of f : Set of all images of elements of A

If S is a subset of A : $f(S) = \{ f(s) \mid s \in S \}$

Functions

Example

See slides § 2.3, p. 2 for the 6th ed. of Rosen's textbook.

Functions

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Problem 3 (p. 99)

Determine whether f is a function from the set of all bit strings to the set of integers if

- $f(s)$ is the number of 1 bits in s

Functions

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Problem 3 (p. 99)

Determine whether f is a function from the set of all bit strings to the set of integers if

- $f(s)$ is the number of 1 bits in s
- $f(s)$ is the position of a 0 bit in s

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Problem 3 (p. 99)

Determine whether f is a function from the set of all bit strings to the set of integers if

- $f(s)$ is the number of 1 bits in s
- $f(s)$ is the position of a 0 bit in s
- $f(s)$ is the smallest integer i such that the i th bit of s is 1 and $f(s) = 0$ when s is the empty string

Functions

Definition

Let A and B be sets. The **Cartesian product** of A and B is

$$A \times B = \{ (a, b) \mid a \in A \wedge b \in B \}.$$

Example

Let $A = \{a, b\}$ and $B = \{1, 2\}$. Then

$$A \times B = \{(a, 1), (a, 2), (b, 1), (b, 2)\}.$$

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Definition

A subset R of the Cartesian product $A \times B$ is called a **relation** from the set A to the set B .

Functions

Definition

Let A and B be sets. A **function** f from A to B is a relation of A to B (i.e., subset of $A \times B$) such that

$$\forall x(x \in A \rightarrow \exists y(y \in B \wedge (x, y) \in f))$$

and

$$\forall x \forall y \forall y' \{((x, y) \in f \wedge (x, y') \in f) \rightarrow y = y'\}.$$

Remark

In some theories different to set theory, the concept of function is a primitive concept.

Injective Functions

Definition

Let $f : A \rightarrow B$. The function f is an **injection** (or **one-to-one**),

if and only if,

$f(a) = f(a')$ implies that $a = a'$ for all $a, a' \in A$.

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if and only if,

$f(a) = f(a')$ implies that $a = a'$ for all $a, a' \in A$.

Example

Whiteboard.

Injective Functions

Definition

Let $f : A \rightarrow B$. The function f is an **injection**, if and only if,

$$\forall x \forall x' (f(x) = f(x') \rightarrow x = x')$$

or equivalent

$$\forall x \forall x' (x \neq x' \rightarrow f(x) \neq f(x')) \quad (\text{contrapositive})$$

where A is the domain of quantification.

Injective Functions

Is a function (non)-injective?

Let $f : A \rightarrow B$.

- To show that f is injective

Show that if $f(x) = f(x')$ for arbitrary $x, x' \in A$ then $x = x'$.

Injective Functions

Is a function (non)-injective?

Let $f : A \rightarrow B$.

- To show that f is injective

Show that if $f(x) = f(x')$ for **arbitrary** $x, x' \in A$ then $x = x'$.

- To show that f is not injective

Find **particular** elements $x, x' \in A$ such that $x \neq x'$ and $f(x) = f(x')$.

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Find **particular** elements $x, x' \in A$ such that $x \neq x'$ and $f(x) = f(x')$.

Exercise

Is the function $f : \mathbb{Z} \rightarrow \mathbb{Z}$, where $f(x) = x^2$ injective?

Surjective Functions

Definition

Let $f : A \rightarrow B$. The function f is a **surjection** (or **onto**),
if and only if,
for every element $b \in B$ there is an element $a \in A$ with $f(a) = b$.

Surjective Functions

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Example

Whiteboard.

Surjective Functions

Definition

Let $f : A \rightarrow B$. The function f is a **surjection**, if and only if,

$$\forall y \exists x (f(x) = y),$$

where A is the domain of quantification for x and B is the domain of quantification for y .

Surjective Functions

Is a function (non)-surjective?

Let $f : A \rightarrow B$.

- To show that f is surjective

Consider an arbitrary element $y \in B$ and find an element $x \in A$ such that $f(x) = y$.

Surjective Functions

Is a function (non)-surjective?

Let $f : A \rightarrow B$.

- To show that f is surjective

Consider an **arbitrary** element $y \in B$ and find an element $x \in A$ such that $f(x) = y$.

- To show that f is not surjective

Find a **particular** $y \in B$ such that $f(x) \neq y$ for all $x \in A$.

Surjective Functions

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Exercise

Is the function $f : \mathbb{Z} \rightarrow \mathbb{Z}$, where $f(x) = x + 1$ surjective?

Bijjective Functions

Definition

A function f is a **bijection** (or **one-to-one correspondence**),
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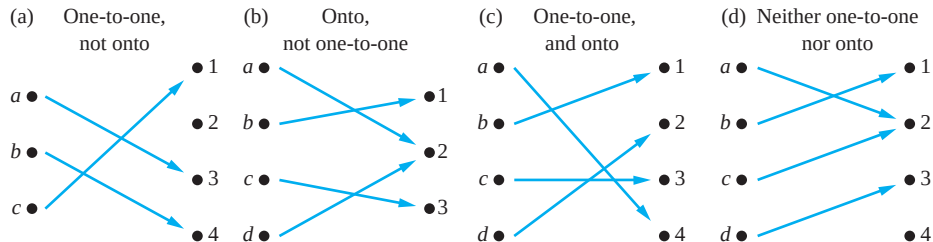
Example (the identity function)

Let A be a set. The identity function $\iota_A : A \rightarrow A$ where $\iota_A(a) = a$ is bijective.

Injective, Surjective and Bijective Functions

Example

Types of functions.[†]



[†]Figure source: (Rosen [1988] 2012, § 2.3, Fig. 5).

Injective, Surjective and Bijective Functions

Problem 16 (p. 100)

Give an example of a function from $\mathbb{N}$ to $\mathbb{N}$:

(a) injective but not surjective

$$f(x) = x + 1.$$

(b) surjective but not injective

$$f(x) = \begin{cases} 0, & \text{if } x = 0 \text{ or } x = 1; \\ x - 1, & \text{otherwise.} \end{cases}$$

(continued on next slide)

Injective, Surjective and Bijective Functions

Problem (continuation)

(c) bijective (but different from the identity function)

$$f(x) = \begin{cases} 0, & \text{if } x = 1; \\ 1, & \text{if } x = 0; \\ x, & \text{otherwise.} \end{cases}$$

(d) neither injective nor surjective

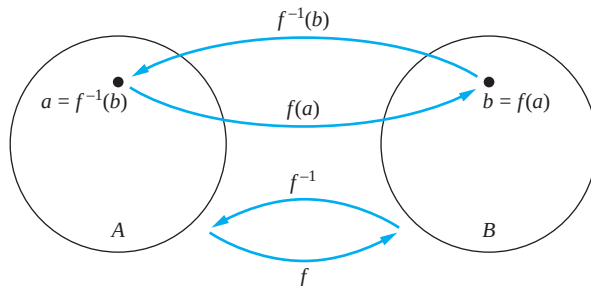
$$f(x) = 5.$$

Inverse Functions

Definition

Let $f : A \rightarrow B$ be a bijective function. The **inverse function** of f , denoted f^{-1} , is the function from B to A defined by[†]

$$f^{-1}(b) = a \text{ iff } f(a) = b.$$



[†]Figure source: (Rosen [1988] 2012, § 2.3, Fig. 6).

Inverse Functions

Exercise

Let $f : \mathbb{Z} \rightarrow \mathbb{Z}$, where $f(x) = x + 1$. Find f^{-1} .

Inverse Functions

Exercise

Let $f : \mathbb{Z} \rightarrow \mathbb{Z}$, where $f(x) = x + 1$. Find f^{-1} .

Question

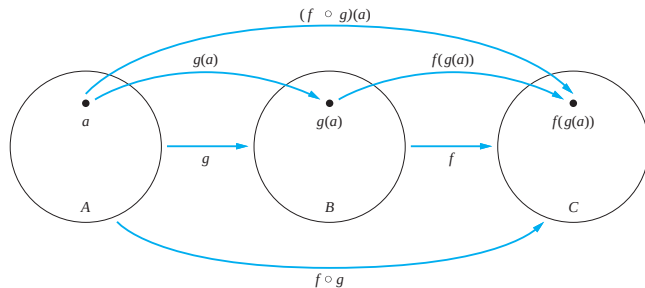
If a function f is not bijective, we cannot define f^{-1} . Why?

Composition of Functions

Definition

Let $g : A \rightarrow B$ and $f : B \rightarrow C$ be two functions. The **composition** of f with g , denoted $f \circ g$, is the function from A to C defined by[†]

$$(f \circ g)(a) = f(g(a)).$$



[†]Figure source: (Rosen [1988] 2012, § 2.3, Fig. 7).

Composition of Functions

Example

See slides § 2.3, p. 8 for the 6th ed. of Rosen's textbook.

Composition of Functions

Example

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Remark

Let f and g be functions. The composition $f \circ g$ cannot be defined unless the range of g is a subset of the domain of f .

Composition of Functions

Example

Let $f : A \rightarrow B$ a bijective function such that $f(a) = b$.

The function $f \circ f^{-1} : B \rightarrow B$ is defined by

$$\begin{aligned}(f \circ f^{-1})(b) &= f(f^{-1}(b)) && \text{(by def. of composition)} \\ &= f(a) && \text{(by def. of inverse function)} \\ &= b && \text{(by def. of inverse function)}\end{aligned}$$

that is, $f \circ f^{-1} = \iota_B$.

Composition of Functions

Remark

In general, the composition of function is not commutative.

Composition of Functions

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In general, the composition of function is not commutative.

Exercise

Let $f, g : \mathbb{N} \rightarrow \mathbb{N}$ be functions, where $f(x) = x^2$ and $g(x) = 2x + 1$. Show that $f \circ g \neq g \circ f$.

Composition of Functions

Problem 25(b) (p. 100)

Let $g : A \rightarrow B$ and $f : B \rightarrow C$ be two functions. Show that if both f and g are surjective functions, then $f \circ g$ is also surjective.

(continued on next slide)

Composition of Functions

Proof

(a) Let $c \in C$.

Composition of Functions

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- (a) Let $c \in C$.
- (b) $f(b) = c$, for some $b \in B$ because f is surjective by hypothesis.

Composition of Functions

Proof

- (a) Let $c \in C$.
- (b) $f(b) = c$, for some $b \in B$ because f is surjective by hypothesis.
- (c) $g(a) = b$, for some $a \in A$ because g is surjective by hypothesis.

Composition of Functions

Proof

(a) Let $c \in C$.

(b) $f(b) = c$, for some $b \in B$ because f is surjective by hypothesis.

(c) $g(a) = b$, for some $a \in A$ because g is surjective by hypothesis.

(d) Then

$$\begin{aligned}(f \circ g)(a) &= f(g(a)) && \text{(by def. of composition)} \\ &= f(b) && \text{(by c)} \\ &= c. && \text{(by b)}\end{aligned}$$

Composition of Functions

Proof

(a) Let $c \in C$.

(b) $f(b) = c$, for some $b \in B$ because f is surjective by hypothesis.

(c) $g(a) = b$, for some $a \in A$ because g is surjective by hypothesis.

(d) Then

$$\begin{aligned}(f \circ g)(a) &= f(g(a)) && \text{(by def. of composition)} \\ &= f(b) && \text{(by c)} \\ &= c. && \text{(by b)}\end{aligned}$$

(e) That is, for all $c \in C$, exists $a \in A$ such that $(f \circ g)(a) = c$.

Composition of Functions

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(a) Let $c \in C$.

(b) $f(b) = c$, for some $b \in B$ because f is surjective by hypothesis.

(c) $g(a) = b$, for some $a \in A$ because g is surjective by hypothesis.

(d) Then

$$\begin{aligned}(f \circ g)(a) &= f(g(a)) && \text{(by def. of composition)} \\ &= f(b) && \text{(by c)} \\ &= c. && \text{(by b)}\end{aligned}$$

(e) That is, for all $c \in C$, exists $a \in A$ such that $(f \circ g)(a) = c$.

(f) Hence, $f \circ g$ is surjective. ■

Composition of Functions

Problem 26 (p. 101)

If f and $f \circ g$ are injections, does it follow that g is injective? Justify your answer.

Composition of Functions

Proof

The function g must be injective.

- (a) Let $g : A \rightarrow B$ and $f : B \rightarrow C$ be two functions, and suppose g is not injective.
- (b) Exists **distinct** elements $x, x' \in A$ such that $g(x) = g(x')$ because g is not injective.
- (c) Then

$$\begin{aligned}(f \circ g)(x) &= f(g(x)) && \text{(by def. de composition)} \\ &= f(g(x')) && \text{(by step 2)} \\ &= (f \circ g)(x'). && \text{(by def. de composition)}\end{aligned}$$

Hence, $f \circ g$ is not injective (**contradiction**).

- (d) Therefore, the function g must be injective. ■

The Graphs of Functions

Definition

Let $f : A \rightarrow B$. The **graph** of f is the set

$$\{ (a, b) \mid a \in A \text{ and } f(a) = b \}.$$

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Example

Whiteboard

The Graphs of Functions

Definition

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$$\{ (a, b) \mid a \in A \text{ and } f(a) = b \}.$$

Example

Whiteboard

Remarks

- Graphs of functions and polymorphic functions

The Graphs of Functions

Definition

Let $f : A \rightarrow B$. The **graph** of f is the set

$$\{ (a, b) \mid a \in A \text{ and } f(a) = b \}.$$

Example

Whiteboard

Remarks

- Graphs of functions and polymorphic functions
- Graphs of functions and programs

The Graphs of Functions

Definition

Let $f : A \rightarrow B$. The **graph** of f is the set

$$\{ (a, b) \mid a \in A \text{ and } f(a) = b \}.$$

Example

Whiteboard

Remarks

- Graphs of functions and polymorphic functions
- Graphs of functions and programs
- Partial and total functions

References

- Kenneth H. Rosen [1988] (2004). *Matemática Discreta y sus Aplicaciones*. Trans. by José Manuel Pérez Morales, Julio Moro Carreño, Ana Isabel Lías Quintero and Pedro Antonio Ramos Alonc. 5th ed. McGraw-Hill (cit. on p. 2).
- [1988] (2012). *Discrete Mathematics and Its Applications*. 7th ed. McGraw-Hill (cit. on pp. 5, 34, 37, 40).